

Olatomiwa Aluko

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EDUCATION

University of Central Florida (UCF)

Orlando, FL

Bachelor of Science in Computer Engineering | GPA: 3.51/4.00

Expected May 2027

- Organizations: Colorstack, National Society of Black Engineers, Knight Hacks
- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Computer Organization, Computer Architecture, Object-Oriented Programming

EXPERIENCE

Bank of New York (BNY)

May 2026 – Aug 2026

Software Engineering Intern

Lake Mary, FL

- Shipped an **8-agent SDLC** review system that detected architecture violations before human review, reducing senior-engineer rework by **2.5 hours per merge request** and saving an estimated **500+ engineering hours**.
- Engineered a **Python** CLI pipeline that used **LLMs** to map **5,000+ vendor financial fields** to BNY's canonical ontology, reducing mapping time **90%+** with parallel execution across **50+ workers**.
- Added **227 automated tests** across **16 modules** and fault isolation to prevent malformed records from halting **500+ field runs**, using only the **Python 3.11+ standard library**.

National Society of Black Engineers – UCF Chapter

Nov 2025 – Present

Founding Software Engineer

Orlando, FL

- Architected and lead a **team of 4 engineers** building a full-stack engagement platform serving **100+ members** across **50+ annual events** with **RBAC** and real-time QR check-in.
- Reduced manual attendance processing **75%** by optimizing **PostgreSQL** queries and caching for **1,000+ concurrent** check-in operations.
- Deployed and maintain a Dockerized **NestJS/Next.js** application across **Railway** and **Vercel**, achieving **99.9% uptime** with Google/Discord **OAuth** and **JWT**-based access control.

PROJECTS

Chalk | *Python, React, FastAPI, PostgreSQL/TimescaleDB, Redis, XGBoost*

- Developed a full-stack NBA statline forecasting platform predicting points, rebounds, assists, three-pointers, steals, blocks, and turnovers, training separate **XGBoost/LightGBM** regressors on **147,000+** historical player-game logs with leakage-safe, time-aware feature engineering.
- Validated forecast accuracy against real completed games across a **1,070-prediction** production sample, achieving a **4.93-point MAE** (meeting quality targets on 5 of 7 tracked stats) benchmarked against Vegas-line accuracy.

Helix | *Go, Python, React, Kubernetes, NATS JetStream, PostgreSQL, ClickHouse*

- Built a distributed agent execution runtime pairing a **Go** control plane with a **Python** worker fleet over **NATS JetStream**, orchestrated on **Kubernetes** with autoscaling and traced end-to-end through **OpenTelemetry**.
- Improved retrieval quality **35%** (Recall@10: **0.253 to 0.343**) by mining hard negatives from agent failure traces to fine-tune a ranking model, validated with bootstrapped confidence intervals.

Aegis | *Go, Adaptive Concurrency Limiting, Circuit Breaking, Prometheus/Grafana*

- Constructed a **Go** reverse proxy protecting streaming backends from **1,500 RPS** traffic spikes using adaptive concurrency limits, priority load shedding, and circuit breakers.
- Delivered **11x more critical video segments (56.0% vs. 4.9%)** than an unprotected backend under a simulated thundering-herd spike, with **1.8ms p50** latency including fast backpressure rejections.

AWARDS

- **4x Hack Winner:** Microsoft x NSBE Agent-a-thon - **1st Place Overall**; LAHacks - **Best Use of Auth0 AI Agents**; HackPrinceton - **Runner-up, Regeneron Best AI & Tech for Clinical Trials**; RenderATL - **Best Use of Snowflake**.

SKILLS

Languages: Python, Java, Go, C/C++, TypeScript, SQL

Frameworks: React, Next.js, FastAPI, NestJS, Jest

Infrastructure & Data: Kubernetes, Docker, PostgreSQL, Redis, NATS JetStream, ClickHouse, OpenTelemetry

Tools & Practices: Git, CI/CD, Prometheus, Grafana, API Design, Unit/Integration Testing, Agile/Scrum